HelpANub Technical Documentation

Made by:

Borja Fernández Ríos

2nd Year of DAM – Multiplatform Application Development

Index

1. System Architecture
2. API
3. Database
4. Web application
5. Mobile application
6. Desktop application
7. Security
8. Deployment
9. User and maintenance guide
10. Future improvements

1. System architecture

The system is composed of:

-Database

-API rest

-Web application

-Mobile application

-Desktop application

Connection system:

1. The web/desktop/mobile application sends a request https to the API.
2. The API process the request, execute the query and returns a JSON, an OK or an internal server error.

Principal technologies used:

-Database: MariaDB

-API rest: Java, Maven

-Web application: PHP, HTML

-Mobile application: Java in Android Studio

-Desktop application: C#, Windows Forms .Net, Control library Windows Forms

2. API

Endpoints list:

Name	Method	Endpoint	Description
subirUsuario	POST	/registro	Creates a new Usuario
accederUsuario	GET	/inicio	Gets a loged user
recogerComentarios	GET	/regogerComentarios	Gets all the comments
recogerRespuestas	GET	/recogerRespuestas	Gets all the answers
darMeGusta	POST	/meGusta	Add mg +1 to the mg of the comment
darNoMeGusta	POST	/noMeGusta	Add nmg +1 to the mg of the comment
subirComentario	POST	/subirComentario	Adds a new comment
subirRespuesta	POST	/subirRespuesta	Adds a new anwer to a comment
eliminarUsuario	POST	/eliminarUsuario	Deletes a user

Https methods:

-POST

-GET

Authentication

The use of a log in with a name and password.

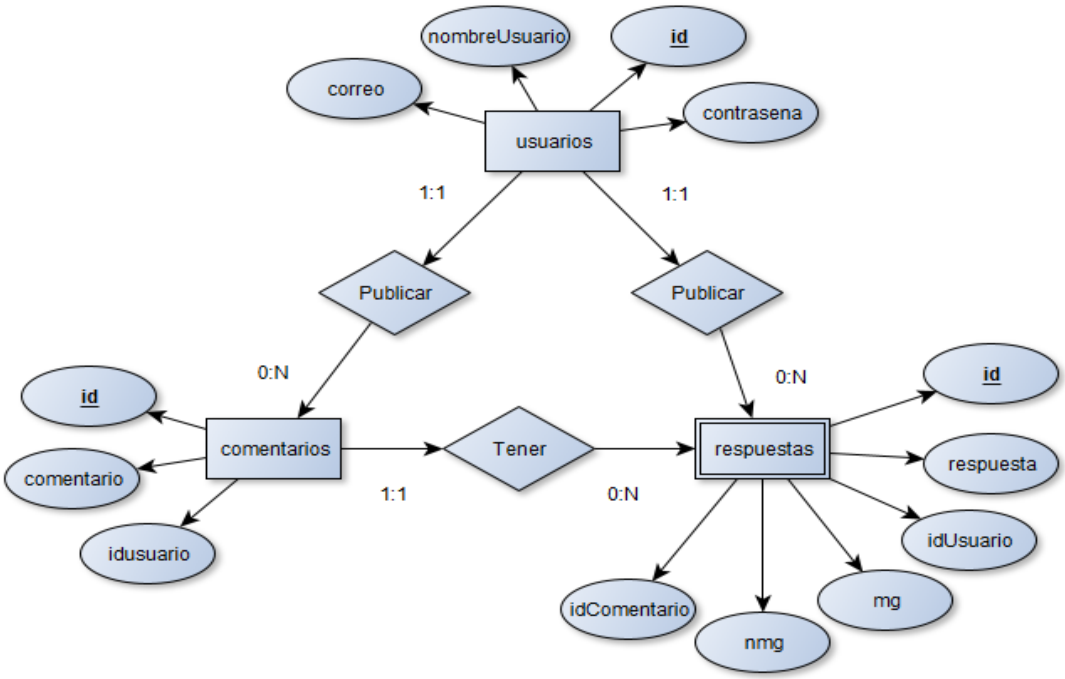
Request examples

Registro:


```
{
  "nombreUsuario": "Borja",
  "correo": "borja@gmail.com",
  "contrasena": "1234"
}
```


3. Database

Entity-Relationship diagram



Description of tables and fields

Usuarios

Name	Type
id	int(15) (Auto_Increment, Primary Key)
nombreUsuario	varchar(40) (Unique)
correo	varchar(100) (Unique)
contrasena	varchar(40)

Comentarios

Name	Type
id	int(15) (Auto_increment, Primary Key)
comentario	varchar(500)
idUsuario	int(15) (FOREIGN KEY (idUsuario) REFERENCES Usuarios(id) ON DELETE CASCADE ON UPDATE CASCADE)

Respuestas

Name	Type
id	int(15) (Auto_increment, Primary Key)
respuesta	varchar(500)
idUsuario	int(15) (FOREIGN KEY (idUsuario) REFERENCES Usuarios(id) ON DELETE CASCADE ON UPDATE CASCADE)
mg	int(50)
nmg	int(50)
idComentario	int(15) (FOREIGN KEY (idComentario) REFERENCES Comentarios(id) ON DELETE CASCADE ON UPDATE CASCADE)

Important rules

-An user can have a lot of comments and answers.

-A comment can only be from one user and a comment.

-A comment can have a lot of answers.

-An answer can only be from one user.

Migrations management

-Importation and exportation from PHPMyAdmin.

4. Web Application

Technologies utilized

-HTML: The structure of the web pages.

-CSS: The style of the web pages.

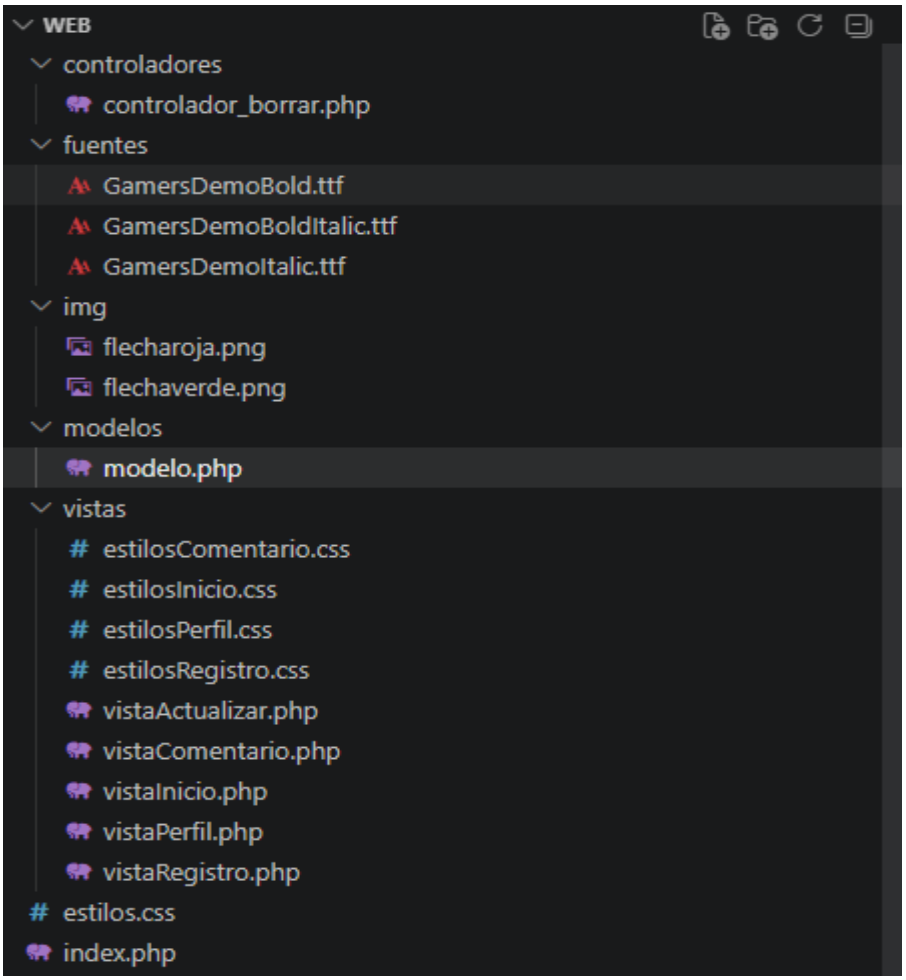
-PHP: The logic of the page.

-CURL: Send HTTP request to the API.

-XAMPP: The use of the database.

-API rest: backend.

Project structure



API Connection

The web connects to the API throught curl:

```
public function __construct()
    {
        $this->baseUrl = "http://localhost:8080/proyecto/rest/proyecto";
    }


    private function request($method, $url, $data = null)
    {
        $ch = curl_init($this->baseUrl . $url);

        curl_setopt($ch, CURLOPT_RETURNTRANSFER, true);
        curl_setopt($ch, CURLOPT_CUSTOMREQUEST, $method);

        if ($data) {
            $jsonData = json_encode($data);
            curl_setopt($ch, CURLOPT_HTTPHEADER, [
                'Content-Type: application/json',
                'Content-Length: ' . strlen($jsonData)
            ]);
            curl_setopt($ch, CURLOPT_POSTFIELDS, $jsonData);
        }

        $response = curl_exec($ch);

        if ($response === false) {
            throw new Exception("Error cURL: " . curl_error($ch));
        }

        $httpCode = curl_getinfo($ch, CURLINFO_HTTP_CODE);
        curl_close($ch);

        if ($httpCode == 404) {
            return null;
        }

        if ($httpCode < 200 || $httpCode >= 300) {
            throw new Exception("Error HTTP: " . $httpCode . " -
Respuesta: " . $response);
        }

        return json_decode($response, true);
    }
```

5. Mobile Application

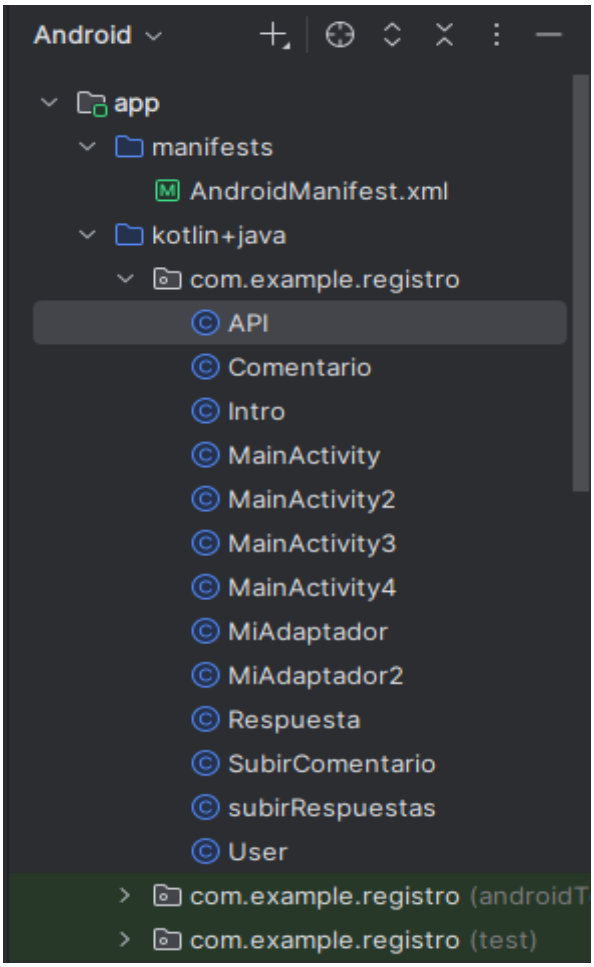
Technologies utilized

-Java and XML in Android Studio.

-HttpURLConnection: to connect the API Rest.

-API Rest: Backend.

Project structure



MainActivity (Login View), MainActivity2 (Home View), MainActivity3 (Registration View), MainActivity4 (Comment View)

API Connection

Connects to the API throught HttpURLConnection (Ex.):

```
public interface RegistroCallback {
    void onRegistro(int idUsuario);
    void onError(String error);
}
public void subirUsuario(String nombre,String correo,String
contrasena,RegistroCallback registroCallback){
    new Thread(() -> {
        try {
            URL url = new
URL("http://10.0.2.2:8080/proyecto/rest/proyecto/registro");
            HttpURLConnection con = (HttpURLConnection)
url.openConnection();
            con.setRequestMethod("POST");
            con.setRequestProperty("Content-Type","application/json");
            con.setDoOutput(true);

            JSONObject json = new JSONObject();
            json.put("nombreUsuario",nombre);
            json.put("correo",correo);
            json.put("contrasena",contrasena);

            System.out.println(json);

            try(OutputStream os = con.getOutputStream()) {

os.write(json.toString().getBytes(StandardCharsets.UTF_8));
            } catch (IOException e) {
                throw new RuntimeException(e);
            }

            int code = con.getResponseCode();
            System.out.println(code);

            if(code == 200 || code == 201){
                BufferedReader reader = new BufferedReader(
                        new InputStreamReader(con.getInputStream(),
StandardCharsets.UTF_8)
                );
                StringBuilder response = new StringBuilder();
                String line;
                while ((line = reader.readLine()) != null){
                    response.append(line.trim());
                }

                JSONObject obj = new JSONObject(response.toString());
                int idUsuario = obj.getInt("id");
                registroCallback.onRegistro(idUsuario);
            } else {
                registroCallback.onError("Error HTTP: " + code);
            }

        }catch (Exception e){
            throw new RuntimeException(e);
        }
    }).start();
}
```

6. Desktop application

Technologies utilized

-C#

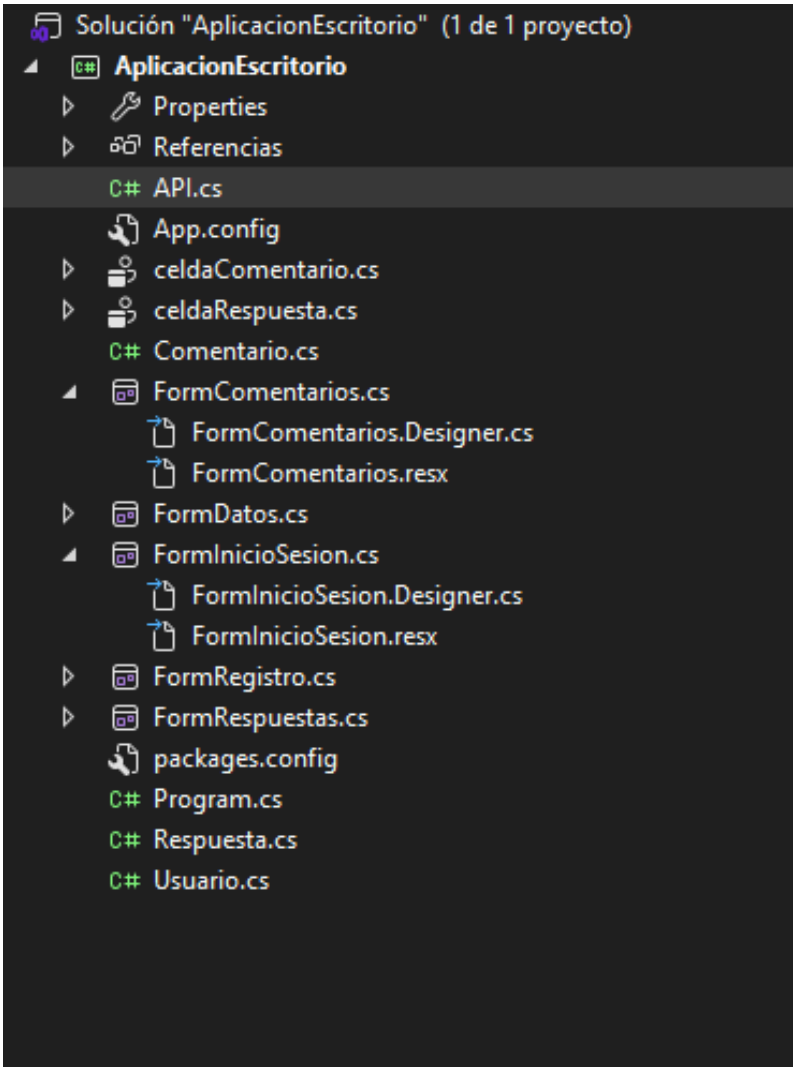
-Windows Forms (.net Framework)

-Windows Forms cotrol library

-HttpClient: Used to send HTTP requests

-JSON: Used to send and receive data from the API Rest

Project structure



API Connection

Connects to the API throught HttpClient and an url (Ex.):

```
public async Task<Usuario> ObtenerUsuario(string nombre, string contra)
{
var response = await
cliente.GetAsync($"{baseUrl}/inicio?nombreUsuario={nombre}&contra={contra}
");

if (response.StatusCode == System.Net.HttpStatusCode.NotFound)
{
    return null;
}

    return await response.Content.ReadFromJsonAsync<Usuario>();
}
```

7. Security

Security mechanisms

-Use of PreparedStatement

User and permission management

-Users can register or log in.

-Users loged can créate comments and answers.

Good Practices

-Use prepared statements to avoid SQL injection.

-Code Organization.

8. Deployment

How to run each application

-Web Application: Open the folder “Web” in Visual Studio Code and execute it with PHP.

-Mobile Application: Open the project in Android Studio and execute the emulator or copy the folder and paste it in the folder “AndroidStudioProjects”.

-Desktop Application: Open the solution .slnx located inside the folder “AplicacionEscritorio”.

API Deployment

1. Open the folder “Proyecto” in Visual Studio Code.
2. Right click on Tombcat and select Add Deployment.
3. Look for a .war named proyecto.war and select it.
4. Star the Tomcat server.

Enviroment

The enviroment is local.

9. Use and maintenance guide

Local execution

1. Open Xampp control panel and start Apache and MySQL.
2. Open the folder that contains the API Rest and start the Tomcat server.
3. Open the web application in Visual Studio Code or desktop application in Visual Studio 2026 or mobile application in Android Studio.

Add features

1. Create a new endpoint in API Rest and do a package to save changes.
2. Update the three applications for the use of that endpoint

Common issues

-Database connection errors.

-Wrong path in the Tomcat server.

-Sdk unknown in Android Studio.

-Start the Tomcat servers and not the xampp control panel.

10. Future improvements

Planned improvements

-When giving mg or nmg on the web appication, the page should update.

-Be able to view the user data in the mobile application.

-Improve the application styles.

-See how many replies each comment has.

New features

-Edit the user loged in web/desktop/mobile application.

-Delete the user loged in desktop/mobile application.

Possible optimizations

-Encrypt users’ passwords.

-Switch from dark to light mode.